

Lab 5 : User and Group Management & Administering Local Password Policies in RHEL 9

Section 1: Understanding Users and Groups

1. **Question:** What is the purpose of users and groups in a Linux system? Provide an example scenario where group permissions are useful.

Answer: To give and strict permissions for specific users in the system,

scenario: when such 1000 user work in company and u want to give them specific permission u can make group attached to the permission and add all user in it , one user can be In more than one secondary group

2. **Question:** Run a command to display a list of all users currently on the system.

Answer: `less /etc/passwd`

3. **Bonus Question:** List all the groups available on the system and identify which group has the most members.

Answer: `less /etc/group | grep ' ; '`

Section 2: Switch to Superuser Account & Grant Superuser Access Using sudo

4. **Question:** How do you switch to the superuser (root) account from a regular user account? Perform this action.

Answer:

```
abdelhamed@eng-abdelhamed:~$ sudo -i
[sudo] password for abdelhamed:
root@eng-abdelhamed:~#
```

5. **Question:** Add an existing user (student) to the sudoers file so they can perform administrative tasks using Sudo. Verify this by switching to the student account and running a command with Sudo.

Answer :

```

root@eng-abdelhamed:~# usermod student -aG sudo
root@eng-abdelhamed:~# su - student
$ id
uid=1001(student) gid=1001(student) groups=1001(student),27(sudo)
$ sudo
usage: sudo -h | -K | -k | -V
usage: sudo -v [-ABknS] [-g group] [-h host] [-p prompt] [-u user]
usage: sudo -l [-ABknS] [-g group] [-h host] [-p prompt] [-U user] [-u user] [command]
usage: sudo [-ABbEHknPS] [-r role] [-t type] [-C num] [-D directory] [-g group] [-h host] [-p prompt] [-R
directory] [-T timeout] [-u user] [VAR=value] [-i|-s] [<command>]
usage: sudo -e [-ABknS] [-r role] [-t type] [-C num] [-D directory] [-g group] [-h host] [-p prompt] [-R
directory] [-T timeout] [-u user] file ...
$ █

```

6. **Bonus Question:** Set up a policy in the /etc/sudoers file that allows the user adminuser to execute any command without being prompted for a password.

```

# User privilege specification
root    ALL=(ALL:ALL) ALL
adminuser ALL=(ALL) NOPASSWD: ALL
# Members of the admin group may gain root privileges
%admin ALL=(ALL) ALL

```

Section 3: Create, Modify, and Delete Local User Accounts

7. **Question:** Create a new user named devopsuser with a home directory and a default shell of /bin/bash.

```

abdelhamed@eng-abdelhamed:~$ sudo useradd -m -s /bin/bash devopsuser
[sudo] password for abdelhamed:
abdelhamed@eng-abdelhamed:~$ █

```

8. **Question:** Modify the user devopsuser to have a different home directory (/home/devuser) and assign them to the group developers.

```

abdelhamed@eng-abdelhamed:~$ sudo usermod devopsuser -d /home/devuser
abdelhamed@eng-abdelhamed:~$ sudo usermod -aG developers devopsuser
abdelhamed@eng-abdelhamed:~$ █

```

9. **Question:** Delete the user devopsuser and remove their home directory.

Answer: `sudo userdel -r devopsuser`

10. **Bonus Question:** View the details of a specific user (e.g., student) using the appropriate command.

Answer:

```
abdelhamed@eng-abdelhamed:~$ finger student
Login: student                      Name:
Directory: /home/student           Shell: /bin/sh
Never logged in.
No mail.
No Plan.
abdelhamed@eng-abdelhamed:~$ grep -i 'student' /etc/passwd
student:x:1001:1001::/home/student:/bin/sh
abdelhamed@eng-abdelhamed:~$
```

Section 4: Create, Modify, and Delete Local Group Accounts

11. **Question:** Create a new group called projectteam.

```
abdelhamed@eng-abdelhamed:~$ sudo groupadd projectteam
abdelhamed@eng-abdelhamed:~$
```

12. **Question:** Add multiple users (student1, student2, student3) to the projectteam group.

```
abdelhamed@eng-abdelhamed:~$ sudo usermod student1 -aG projectteam
abdelhamed@eng-abdelhamed:~$ sudo usermod student2 -aG projectteam
abdelhamed@eng-abdelhamed:~$ sudo usermod student3 -aG projectteam
abdelhamed@eng-abdelhamed:~$
```

13. **Question:** Modify the projectteam group by changing its group ID (GID) to 1050.

```
abdelhamed@eng-abdelhamed:~$ sudo groupmod projectteam -g 1050
```

14.Question: Delete the group projectteam.

Answer :

```
abdelhamed@eng-abdelhamed:~$ sudo groupdel -f projectteam
abdelhamed@eng-abdelhamed:~$
```

Section 5: Set a Password Management Policy for Users

15. **Question:** Set the password expiry for the user student1 to expire in 30 days using the change command.

```
abdelhamed@eng-abdelhamed:~$ sudo chage -M 30 student1
abdelhamed@eng-abdelhamed:~$
```

16. **Question:** Force a password change for the user student2 at the next login.

```
abdelhamed@eng-abdelhamed:~$ sudo passwd -e student2
passwd: password expiry information changed.
abdelhamed@eng-abdelhamed:~$
```

17. **Question:** Display the password aging details for the user student3.

```
abdelhamed@eng-abdelhamed:~$ sudo chage -l student3
Last password change                : Dec 08, 2025
Password expires                    : never
Password inactive                    : never
Account expires                     : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
abdelhamed@eng-abdelhamed:~$
```

18. **Bonus Question:** Set a global password policy that requires passwords to be at least 10 characters long.

Section 6: Lock and Unlock User Accounts

19. **Question:** Lock the account of user student3 to prevent them from logging in.

```
abdelhamed@eng-abdelhamed:~$ sudo usermod -L student3
abdelhamed@eng-abdelhamed:~$
```

20. **Question:** Unlock the account of student3 to allow them to log in again.

Answer : first u have to put password first before unlocking the account

```
abdelhamed@eng-abdelhamed:~$ sudo usermod -U student3
abdelhamed@eng-abdelhamed:~$
```

21. **Bonus Question:** What is the difference between locking a user's account and disabling their password?

Locking the user account : make the user cannot login to its account through ssh or using password in the TTY , the all account services and locked and not working

Disabling the password : u can't login to the account using the password , put the account service still work with services don't need password operations on it